Alex Hong

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EDUCATION

Stanford University — B.S. Mechanical Engineering · GPA 3.7 EXPECTED JUNE 2028

Coursework: Thermodynamics, Fluid Mechanics, Dynamics, ME 102 Product Realization, ME 103, Linear Algebra, Multivariable Calculus, Differential Equations.

RESEARCH

Summer Undergraduate Researcher, Boies Lab — Stanford School of Engineering JUNE – SEPT 2026

- Compared counterflow and main-flow injection in a 1300 °C floating-catalyst CVD reactor; counterflow raised carbon nanotube production to 216 mg/hr. Found the optimal injection point, 28.3 cm, through repeated runs with a PhD collaborator.
- Designed quartz-tube flange assemblies in Fusion 360, including sealing geometry and injector-tube integration for high-temperature service.

ENGINEERING PROJECTS

Custom Aluminum Piano — ME 103 FALL 2026 · IN PROGRESS

- 20-key MIDI keyboard made entirely of 356A aluminum; frame, base, and key blanks sand cast in the PRL foundry from 3D-printed PLA patterns, then finished on a manual mill.
- Wrote the operations sequence (.060 machining stock, 2° draft, ×1.013 shrink; all 20 pivot holes drilled in one setup) and found five CAD fixes before pattern-making.
- Built the BOM from CAD pour volumes: $104.84 estimated against a $130 budget.

Spring-Powered Ping Pong Launcher — ME 102 Product Realization WINTER 2026

- Designed and built a purely mechanical spring launcher in a 12 × 12 × 24 in envelope, firing through a target at 5–7 ft; cut the static counterweight, the primary scored metric, through geometry instead of added mass.
- Produced free-body diagrams, force/moment and stored-energy calculations, and a dimensioned drawing package with tolerances, BOM, and exploded views; parts laser-cut and 3D-printed.

Outdoor Stage Build — Sigma Nu Sunsplash 2026

- Designed a wooden stage, DJ table, and tiered platforms (17 platforms, 8 fence sections) in Fusion 360 for an event with ~1,340 RSVPs; sized 2×4, 4×4, and plywood members, then built it on site with a group. The aluminum truss was purchased. DJ'd the event from the stage.

EXPERIENCE

Tech Desk Consultant — Stanford University SEPT 2025 – PRESENT

- Resolve hardware and software issues for students and staff; manage equipment checkout and large-format printing (6 hrs/week).

Sales Intern — Your Energy Solutions, Livermore, CA JUNE – SEPT 2025

- Learned residential and commercial PV system design, from panel and inverter specs through per-customer ROI and NEM 3.0 incentives; generated 50+ qualified commercial leads in two months.

Machine Learning Intern — Lento Medical, Katy, TX JUNE – AUG 2023

- Operated CNC machines producing patient-specific knee implant components; researched machine learning methods for X-ray analysis and surgical planning for the PtoleKnee implant system.

SKILLS

EXPERIMENTAL	High-temperature reactor operation, CVD process iteration, design of experiments
ANALYSIS	Free-body diagrams, force/moment balance, energy analysis, dimensional analysis
CAD & FABRICATION	Fusion 360, engineering drawings, dimensioning and tolerancing, sand casting and pattern design, manual milling, operations planning, 3D printing, laser cutting, CNC
SOFTWARE	Python
LANGUAGES	English (native), Korean (fluent)

ADDITIONAL

Producer and DJ as Hongsta; self-released album *Reaching for the Moon* (Aug 2026); opened for Poolside and Twinsick. Lead trumpet: Carnegie Hall with NYO Jazz (2024), European tour under Luis Bonilla (2025), Clark Terry Big Band under Wynton Marsalis (2023). Interests: motorsport, soccer, chess, grilling.